Practical 5

Write a program to implement Shortest Job First (SJF) Preemptive Scheduling for three processes and calculate the total context switches and average waiting time. The processes have burst times 10ns, 20ns, and 30ns, arriving at 0ns, 2ns, and 6ns, respectively.

~/process

```
GNU nano 8.7                                                    srtf.c
t main() {

  int n = 3;

  int at[] = {0, 2, 6};       // Arrival Time
  int bt[] = {10, 20, 30};    // Burst Time
  int ct[] = {10, 30, 60};    // Completion Time (from SRTF)

  int wt[3], total_wt = 0;

  for (int i = 0; i < n; i++) {
      wt[i] = ct[i] - at[i] - bt[i];
      total_wt += wt[i];
  }

  printf("Process\tArrival\tBurst\tWaiting\n");
  for (int i = 0; i < n; i++) {
      printf("P%d\t\t%d\t\t%d\t\t%d\n", i+1, at[i], bt[i], wt[i]);
  }

  printf("\nAverage Waiting Time = %.2f ns\n",
         (float) total_wt / n);

  printf("Total Context Switches = 2\n");

  return 0;

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```

```
lenovo@LAPTOP-6627EU0U MSYS ~/process
$ ./srtf
Process Arrival Burst   Waiting
P1              0               10              0
P2              2               20              8
P3              6               30              24

Average Waiting Time = 10.67 ns
Total Context Switches = 2
```